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Taxi Demand Forecasting

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Group 7

</Project Overview

This project focuses on:

- Developing time series regression models
- Forecasting taxi demand
- Applying Scrum methodology
- Using Jira to simulate Agile workflow
- Completing 2 full sprints (1 week each)



</Problem Definition & Business Value

Technical Problem:

Forecast taxi demand using time series regression.

- Predict hourly/daily taxi demand
- Use historical ride data
- Apply regression / ML and DL models
- Evaluate prediction accuracy



Business Problem:

Mismatch between rider demand and driver supply.

Problems caused by mismatch:

- Long rider wait times
- Driver idle time
- Lost revenue
- Surge pricing inefficiencies



</Problem Definition & Business Value



Business Value:

- Increase Revenue
- Reduce wait time
- Improve driver utilization
- Optimize pricing
- Enable strategic planning



</Agile Team Roles



Yuvraj Handa

Development Team Member



Kopinath
Sithamparanathan

Development Team



Antoine Saldagna

Product Owner



Elizaveta Khoroshilova

Scrum-master



Daria Ignateva

Development Team Member



Hari Saravanan

Development Team



Nithin Reddy

Development Team

</Project Goal & Scope

Build a time series regression model that predicts taxi demand accurately enough to support operational decisions.

Scope:

- Data collection
- Data preprocessing
- Feature engineering
- Model development
- Evaluation
- Agile documentation

Development Team KPI: Achieve an RMSE of **strictly** < 40 orders per hour.

</Success Criteria

The project is considered successful if:



Model achieves acceptable MAE / RMSE



Forecast trend aligns with real demand



Two complete Scrum cycles are documented



All Scrum artifacts are included

</Stakeholders & Users

Stakeholders:

- Taxi Company Management
- Operations Team
- Pricing Team

End Users:

- Dispatch system
- Drivers
- Business analysts

Stakeholder Needs:

- Accurate demand forecasts

</Timeline

Sun	Mon	Tue	Wed	Thu	Fri	Sat
1	2	3	4	5	6	7
8	9	10	11	12	13	14
15						
	16	17	18	19	20	21
22	23	24	25	26	27	28
29	30					

</Sprint 1

</Sprint 2

</Meetings

</Meeting Agenda

	Mon	Tue	Wed	Thu	Fri
Week 1	Define data prep goals and assign roles.		Discuss feature engineering and resampling progress.		Baseline review & improvement planning
Week 2		Distribute ML models (CatBoost, RF, LR, XGBoost) to team.	Hyperparameter tuning updates and NaN bug resolution.	Evaluate model metrics, finalizing	

</Sprint 1

</Sprint 2

</Sprint 1: Data & Feature Engineering



Data Cleaning &
Resampling

{01}



Feature
Engineering

{02}



Data Split

{03}

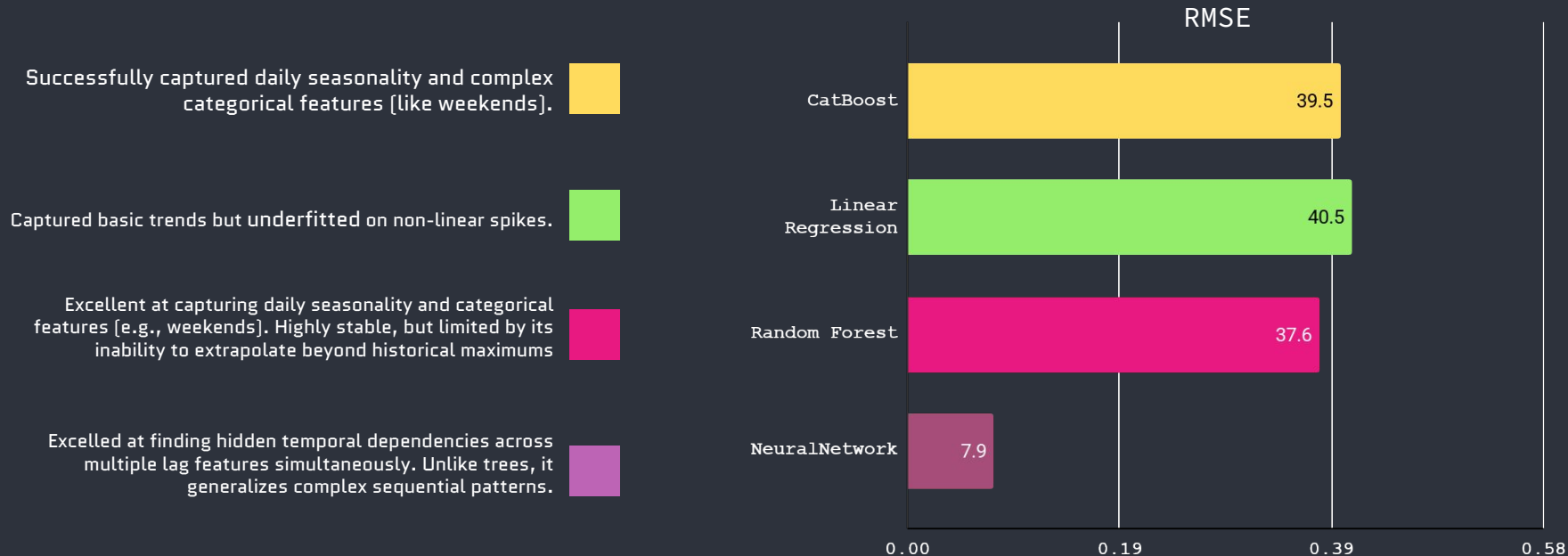


Baseline model

{04}



</Sprint 2 — ML Modeling Competition



Scrum Artifacts

Bug: Minor Overfitting



Description

- Observed training RMSE significantly lower than validation RMSE
- Mitigated using early stopping and regularization

HS

Hari Saravanan
February 25, 2026 at 10:21 PM

Got it Liza. This will take about a day to complete

EK

Elizaveta Khoroshilova
February 25, 2026 at 9:25 PM

Hi Hari! Please take the Random Forest model. We need to beat the baseline (58) and ideally get RMSE under 40. Good luck!

Spaces

AASD_4013_TaxiPrediction

Summary Backlog Board Calendar Timeline Pages Forms Reports

Search board Filter

TO DO 2 OF 2

Task 3: EDA and Resampling

SCRUM-9

Task 4: Feature Engineering

SCRUM-13

+ Create

IN PROGRESS 2 OF 2

Evaluate Model Performance

SCRUM-16

Predict Next-Hour Taxi Demand

SCRUM-17

IN REVIEW 6 OF 6

Task 2: Data preparation

SCRUM-5

Model 3: Catboost

SCRUM-21

Model 2: Random Forest

SCRUM-18

Evaluate Random Forest Model Performance

SCRUM-19

Predict Next-Hour Taxi Demand using Random Forest

SCRUM-20

Train a Neural Network

SCRUM-28

DONE 2 OF 2

Task 1.1: Setup Agile Board and Roles

Feb 27, 2026

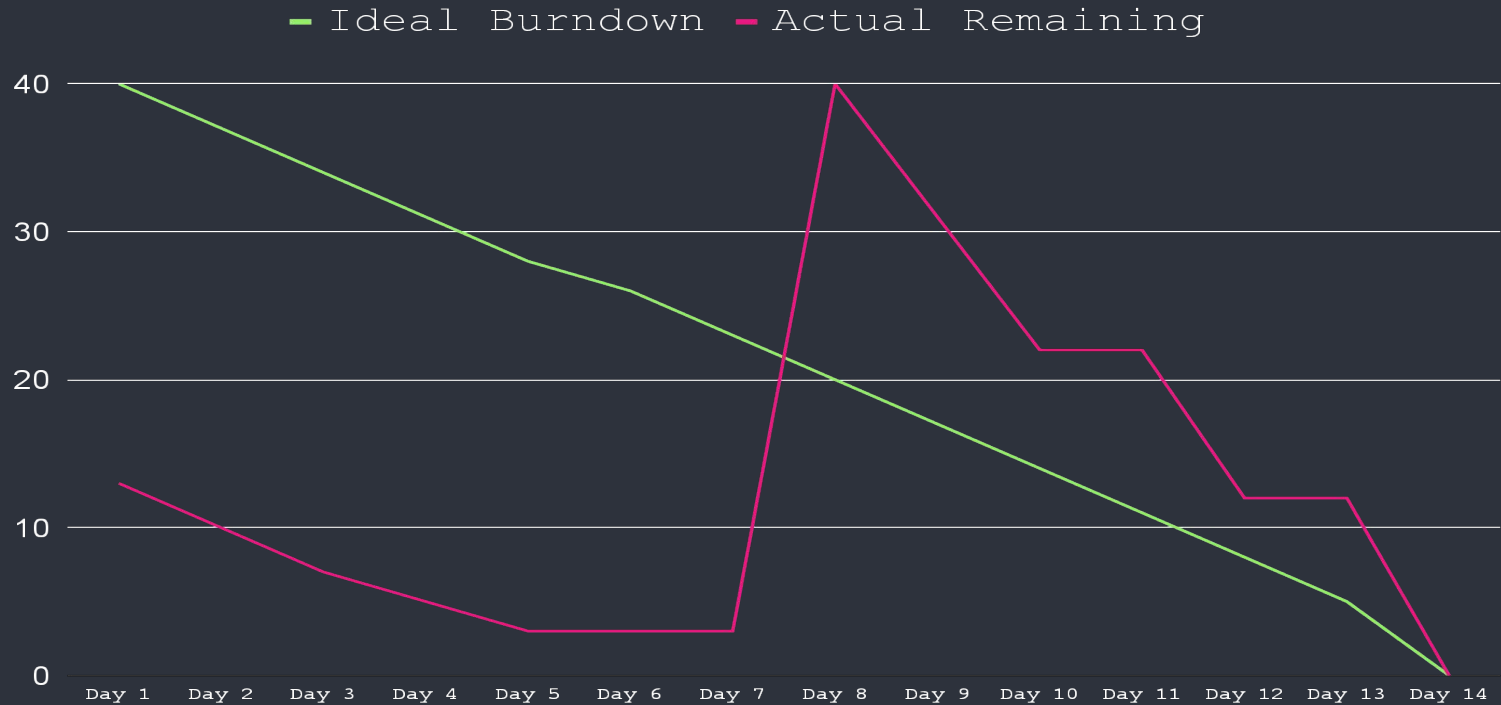
SCRUM-1

Model 1: Train Linear Regression Model

SCRUM-15

13

Burndown Chart



</Lessons learned



Iterative Development Works Better



Early evaluation enabled adaptation



Improved planning and workload balancing is needed



Regular communication reduced blockers